

Arko Bera

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EDUCATION

Indian Institute of Information Technology, Nagpur
B.Tech in Data Science and Analytics

2023 – 2027
CGPA: 8.86

TECHNICAL SKILLS

Programming & Data: Python, SQL, NumPy, Pandas, Scikit-learn

Deep Learning: PyTorch, TensorFlow, CNNs, RNNs, Transformers, Hugging Face, LangGraph, LangChain, LangSmith

MLOps & Cloud: MLflow, DVC, DagsHub, Docker, AWS (S3, EC2), Grafana

Tools: Git, GitHub, VS Code, Jupyter, Streamlit, Gradio

Domains: NLP, Computer Vision, LLMs, Audio ML, Model Optimization

EXPERIENCE

Open Source Projects — Kaggle

Jul 2024 – Present

- **Charting Student Math Misunderstanding** — Built a multi-model NLP pipeline to identify student misconceptions from incorrect answers by using **Parameter-Efficient fine-tuning (PEFT)** on **DeepSeek Math 7B**, **Hunyuan 7B**, and **Qwen 4B**. Ranked **453 / 1,800+** participants.
- **PhysioNet EEG Digitization** — Designed a **three-stage computer vision pipeline** to convert EEG waveform images into digital signals: (1) image orientation correction, (2) preprocessing and noise reduction, and (3) signal extraction. Improved **Signal-to-Noise Ratio (SNR)** to **18.14** and secured **Rank 161 / 1,421** participants.

PROJECTS

Accident Risk Prediction System — MLOps, AWS, MLflow, Flask, FastAPI

- Designed a **production-grade MLOps pipeline** using **DVC** for versioning and **MLflow** to track 20+ experiments.
- Integrated **AWS (S3, ECR, EC2)** with **Docker** and **GitHub Actions** for automated CI/CD and deployment.
- Built a **FastAPI backend** for real-time inference and a **Flask frontend** for user interaction.
- **Result:** Minimized **RMSE to 0.05544**, reducing retraining time from hours to minutes, and enabled sub-second latency.

Emotion-Aware Text-to-Speech System — LLMs, Transformers, TTS, FastAPI, Streamlit

- **Problem:** Traditional TTS systems produce **emotionless, robotic speech** lacking contextual expressiveness.
- **Approach:** Combined **transformer-based emotion detection** with **voice parameter mapping (pitch: +4st / -3st, rate: 1.1x / 0.9x)** and **SSML-based modulation** in a multi-provider TTS pipeline.
- **Result:** Delivered **real-time, emotion-aware speech synthesis** with **parameter-controlled voice modulation (pitch, rate)** and multi-provider fallback (Google → ElevenLabs → gTTS).

Physics-Informed Neural Networks for Fluid Flow — PINNs, OpenFOAM, CFD, PyTorch

- **Problem:** Traditional CFD methods are **computationally expensive and mesh-dependent**, limiting scalability to complex **2D/3D** geometries.
- **Approach:** Created **PINN and FPINN models** to solve **incompressible Navier–Stokes equations** using **physics-constrained loss** (data + PDE) and validated against **OpenFOAM simulations** across multiple geometries.
- **Result:** Reduced prediction error in flow fields (**MAE(u)=0.0138**, **3D MAE(p)=0.006**), demonstrating **mesh-free CFD approximation** with improved performance using **Fourier feature embeddings**.

ACHIEVEMENTS

- **Global Rank 203** in **Amazon ML Challenge 2025** (7,300+ teams).
- Solved **600+** algorithmic problems on LeetCode (Max Rating: **1876**).
- **Ranked 8** in **PS5E4, PS6E3** on Kaggle.